

Reasoning and translating to First-Order Logic using Large Language Models on FOLIO^[1]

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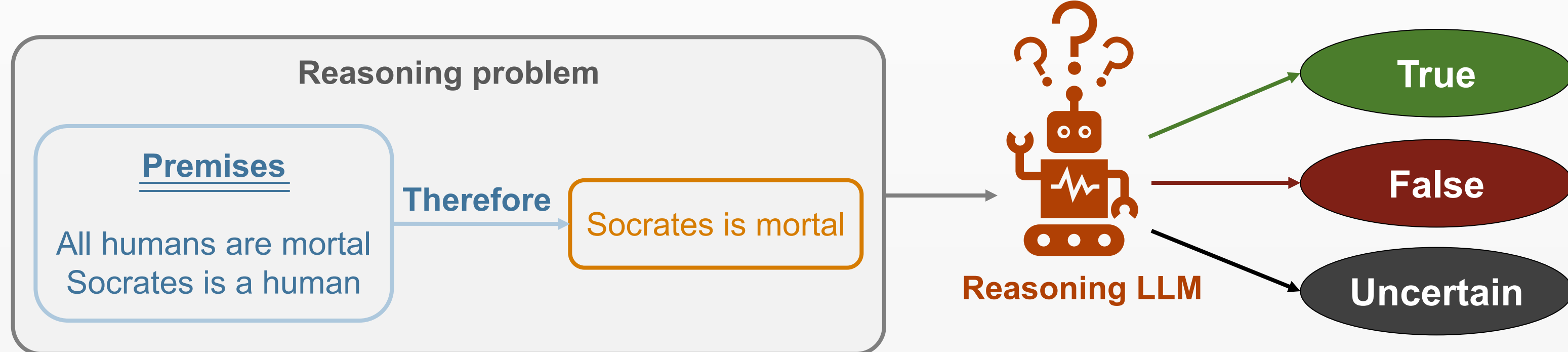
Introduction

Motivation

- First-Order Logic (FOL) is widely used for symbolic reasoning and logical entailment.
- Recent LLMs enable reasoning and translating natural language into FOL.

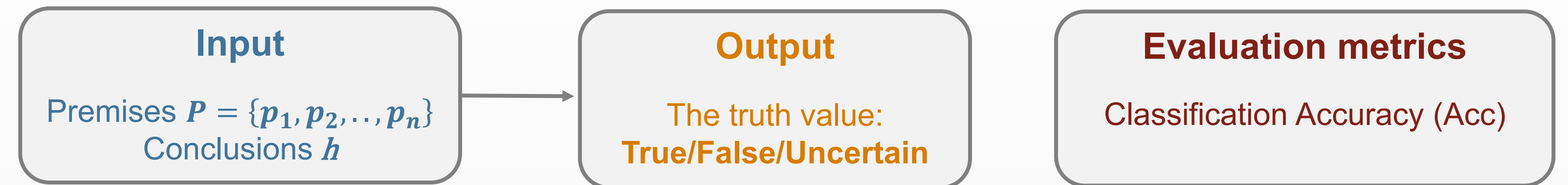
Research Gap

LLMs lack a robust mechanism for FOL reasoning, and current translations often introduce syntax errors and fail to preserve semantic meaning.

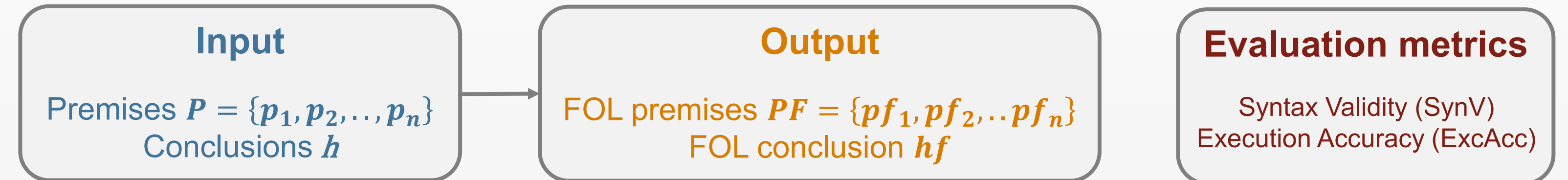


Problem formulation

Problem 1 – Natural language reasoning with first-order logic

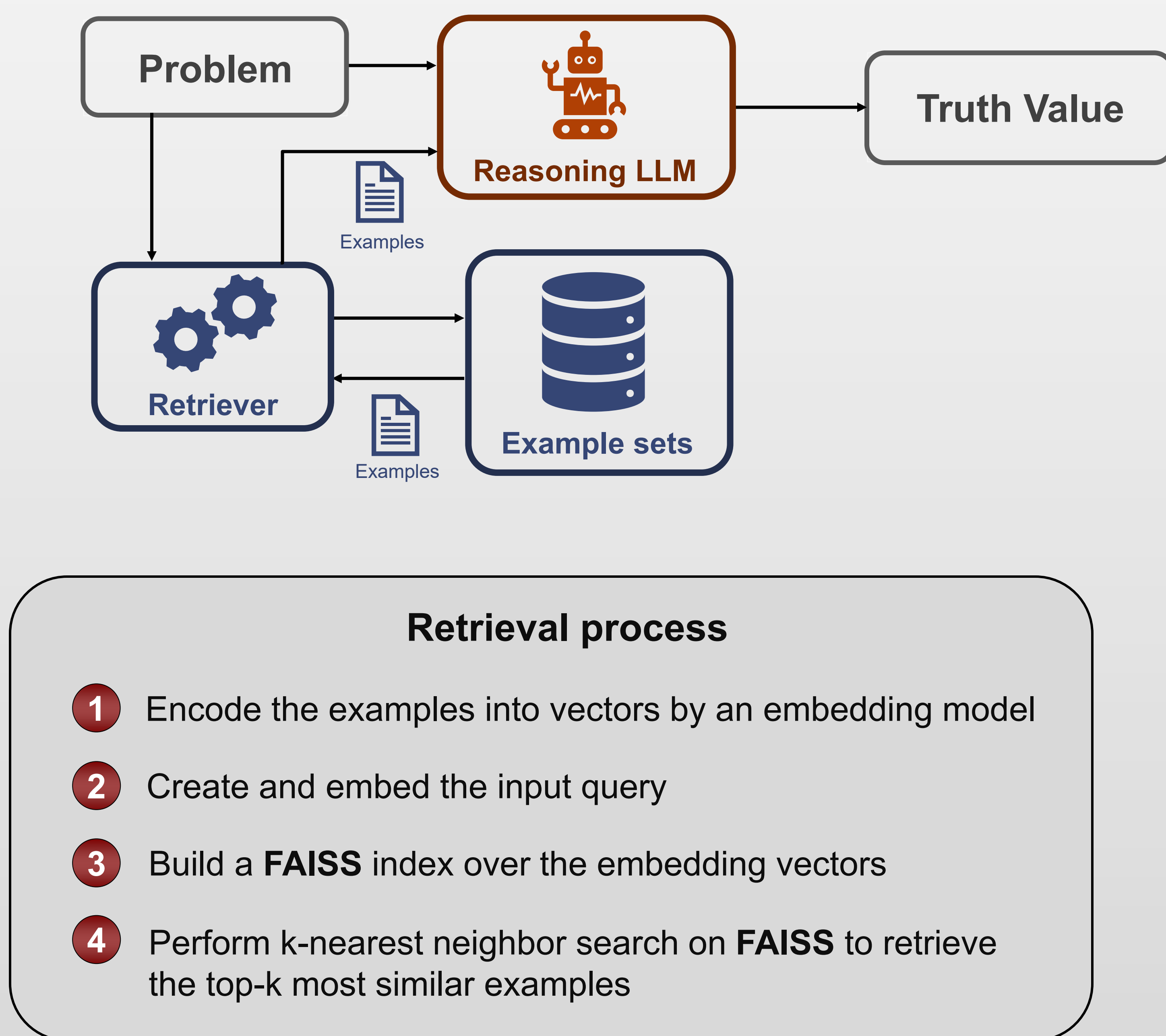


Problem 2 – NL-to-FOL translation

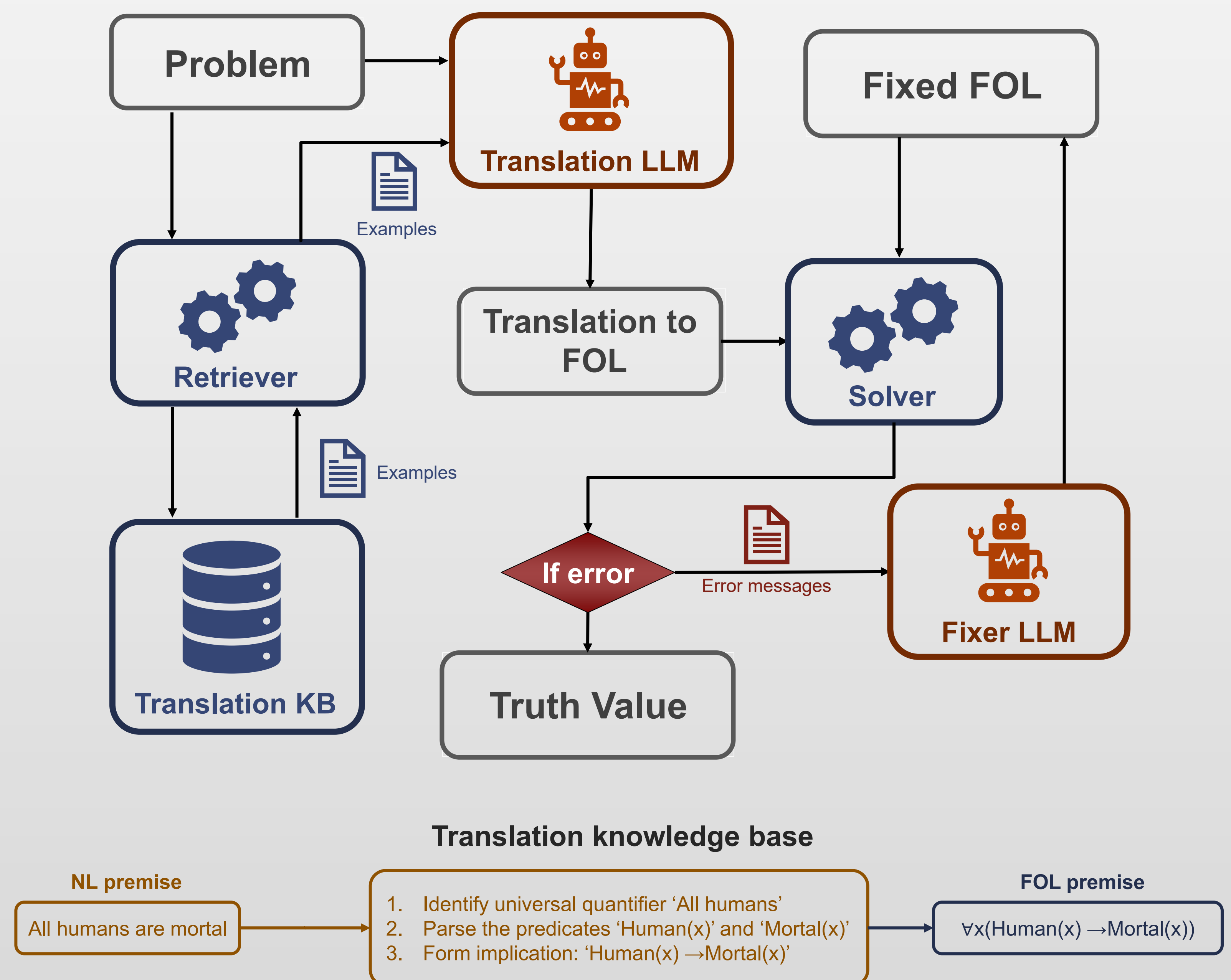


Methodology

Method 1 – Semantic RAG (SeRAG)



Method 2 – Step-by-step Translation based on RAG (ST-RAG)*



*This work is partially inspired by [2]

Experimental results

Performance evaluation on Logical reasoning

Algorithm	Model	Acc(%)
LLM reasoning	Qwen2.5:3b	36.27
LLM reasoning	Qwen2.5:14b	52.94
LoT [3]	Llama3.1	35.30
LoT [3]	Ministral-3:8b	42.60
LoT [3]	Qwen2.5:14b	58.30
SeRAG (Ours)	Qwen2.5:14b	63.73
ST-RAG (Ours)	Qwen2.5:14b	64.22

SeRAG and ST-RAG have improved the classification accuracy over other LLM baselines, showing that retrieved logical examples help models choose more correct truth values.

Performance evaluation on NL-FOL translation

Algorithm	Model	SynV(%)	ExcAcc(%)
LLM translation	Qwen2.5:3b	59.31	42.17
LLM translation	Qwen2.5:14b	71.08	62.07
ST-noRAG	Qwen2.5:14b	83.33	65.29
ST-RAG w/o Fixer	Qwen2.5:14b	78.92	68.94
ST-RAG (Ours)	Qwen2.5:14b	93.13	66.32

ST-RAG has improved the syntax validity and the execution accuracy compared to plain LLM translation, confirming the benefits of retrieving step-by-step translation examples, solver feedback, and the Fixer LLM.

Conclusion

This work presents two frameworks: (i) SeRAG, a framework for logical reasoning over FOL, and (ii) ST-RAG, which addresses the NL-to-FOL translation problem by combining RAG with a step-by-step translation knowledge base and a syntax correction module. Both methods substantially improve the evaluation metrics on the FOLIO dataset. However, these pipelines still depend on a manually curated translation-example knowledge base, and the model can struggle with long inputs and deeper semantic errors that are not easily corrected.

Future work will focus on

- Model design improvement:** Expanding and automatically constructing the knowledge bases, improving error correction.
- Model finetuning:** Applying finetuning techniques such as LoRA for logical reasoning task.
- Evaluation metrics:** Exploring or designing more reliable metrics for NL-FOL translation problem.

References

- [1] HAN, Simeng, SCHOELKOPF, Hailey, ZHAO, Yilun, et al. Folio: Natural language reasoning with first-order logic. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 2024.
- [2] Liu, Tongxuan, et al. "Logic-of-thought: Injecting logic into contexts for full reasoning in large language models." *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*. 2025.
- [3] Hu, Ruikang, et al. "LTRAG: Enhancing autoformalization and self-refinement for logical reasoning with thought-guided RAG." *Findings of the Association for Computational Linguistics: ACL 2025*. 2025.